

From p_0, p_1, p_2 to Lepton Masses

ϕ -Skeleton and Galois Correction Factors

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From p_0, p_1, p_2 to Lepton Masses: ϕ -Skeleton and Galois Correction Factors

Abstract

Doc. 293 provides the icosahedral redistribution weights $p_0 = 2/9$, $p_1 = (2+3\phi)/9$, $p_2 = (5-3\phi)/9$. Doc. 367 shows that $p_0 = \theta$ and rewrites the Koide formula entirely in matrix elements of R_5 – the mass bridge via θ is thereby obtained directly from the A_5 geometry.

This document investigates a second, purely algebraic route: the ratios p_1/p_2 and p_0/p_2 carry the ϕ -skeleton of the lepton masses directly, without a cosine function. Main result: $p_1/p_2 = \phi^8$ and $p_0/p_2 = 2\phi^4$ are exact algebraic identities **[B]**. The mass ratios m_τ/m_e and m_μ/m_e follow approximately by multiplication with the Galois factors 74 and 15 from Doc. 338 (accuracy 0.03 % and 0.56 %) **[K]**. The derivation of these Galois factors from the $\mathbb{Z}_3$ framework remains open **[S]**.

Notation

Epistemic markers:

- [B]** algebraically proved, exact arithmetic
[K] numerically confirmed (approximation, accuracy stated)
[S] structurally observed, derivation open
[E] established external mathematics

Symbol	Meaning
ϕ	golden ratio $(1 + \sqrt{5})/2$
p_0, p_1, p_2	icosahedral redistribution weights from Doc. 293
θ	Koide phase; $\theta = p_0 = 2/9$ (Doc. 367)
a_k	Koide amplitude: $\sqrt{m_k} = M(1 + \sqrt{2} \cos(\theta + 2\pi k/3))$

1 Starting Point

Doc. 293 computes the redistribution weights of the electron mode $\nu_e = \nu_1$ under the icosahedral five-fold rotation $R_5 \in A_5$:

$$p_0 = \frac{2}{9}, \quad p_1 = \frac{2+3\phi}{9}, \quad p_2 = \frac{5-3\phi}{9}, \quad p_0 + p_1 + p_2 = 1.$$

Doc. 367 establishes that $\theta = p_0 = 2/9$: the same quotient appears once as Koide phase and once as transition probability. Doc. 367 has shown that the mass bridge via $\theta = p_0$ can be obtained directly from the A_5 geometry. The question of this document is a different one: do the *ratios* p_1/p_2 and p_0/p_2 carry the ϕ -skeleton of the masses algebraically – without cosine function, without θ as a running variable?

2 Exact Algebraic Identities

$$p_1/p_2 = \phi^8$$

$$\frac{p_1}{p_2} = \frac{2+3\phi}{5-3\phi} = \frac{(7+3\sqrt{5})/2}{(7-3\sqrt{5})/2} = \frac{(7+3\sqrt{5})^2}{49-45} = \frac{94+42\sqrt{5}}{4} = \frac{47+21\sqrt{5}}{2}. \quad (1)$$

On the other hand: $\phi^4 = 3\phi + 2$ (from $\phi^2 = \phi + 1$), so $\phi^8 = (3\phi + 2)^2 = 21\phi + 13 = \frac{47+21\sqrt{5}}{2}$. Hence **[B]**:

$$\boxed{p_1/p_2 = \phi^8.}$$

$$p_0/p_2 = 2\phi^4$$

$$\frac{p_0}{p_2} = \frac{2}{5-3\phi} = \frac{2(7+3\sqrt{5})}{4} = \frac{7+3\sqrt{5}}{2}.$$

And $2\phi^4 = 6\phi + 4 = 7 + 3\sqrt{5}$. Hence **[B]**:

$$\boxed{p_0/p_2 = 2\phi^4.}$$

$$p_1/p_0 = \phi^4/2$$

Direct consequence: $p_1/p_0 = \phi^8/(2\phi^4) = \phi^4/2$ **[B]**. All three ratios carry only ϕ -powers with rational prefactors.

Amplitude Structure

The square roots $3\sqrt{p_j}$ (amplitudes up to normalisation factor 3) take exactly the icosahedral values **[B]**:

$$3\sqrt{p_0} = \sqrt{2}, \quad 3\sqrt{p_1} = \phi^2, \quad 3\sqrt{p_2} = \frac{1}{\phi^2}.$$

3 Approximate Connection to Mass Ratios

Precision limit: $\cos(2/9)$ is transcendental

The exact Koide mass ratios at $\theta = 2/9$ are:

$$\frac{m_\tau}{m_e} = \left(\frac{a_0}{a_1}\right)^2, \quad \frac{m_\mu}{m_e} = \left(\frac{a_2}{a_1}\right)^2,$$

with $a_k = 1 + \sqrt{2}\cos(2/9 + 2\pi k/3)$. Since $2/9 \notin \pi\mathbb{Q}$, $\cos(2/9)$ is transcendental **[E]**. Consequently m_τ/m_e and m_μ/m_e cannot be expressed exactly as rational multiples of ϕ -powers. All connections below are approximate identities with stated accuracy.

$$m_\tau/m_e \approx 74 \times \phi^8 = 74 \times p_1/p_2$$

Numerically (with $\theta = 2/9$ exact in the Koide formula) **[K]**:

$$\frac{m_\tau}{m_e}\bigg|_{\theta=2/9} = 3477.47, \tag{2}$$

$$74 \times \phi^8 = 3476.42, \tag{3}$$

$$\text{deviation} = 0.030 \%. \tag{4}$$

The Galois factor $74 = 2 \times 37$, where $37 \mid |GF(3^{18})^*|$, appears in Doc. 338 as part of the correction factor $K_{\text{frak}} = 74/75$ of the α derivation. Its appearance here is a structural observation **[S]**; the derivation from the $\mathbb{Z}_3$ framework is outstanding.

$$m_\mu/m_e \approx 30 \times \phi^4 = 15 \times p_0/p_2$$

$$\frac{m_\mu}{m_e}\bigg|_{\theta=2/9} = 206.770, \tag{5}$$

$$30 \times \phi^4 = 205.623, \tag{6}$$

$$\text{deviation} = 0.56 \%. \tag{7}$$

Here the approximation is considerably less precise. The factor $15 = 3 \times 5$, with $3 = |GF(3)|$ and $5 = \min.\text{prim}(|GF(81)^*|)$, comes from the Galois hierarchy of Doc. 338 **[S]**.

4 Numerical Confirmation

The check script `pruef_368_p_massen.py` (2/python/Dok368_Skripte) confirms **[K]**:

1. $p_1/p_2 = \phi^8$ exactly (dev. $< 10^{-35}$, 40 digits).
2. $p_0/p_2 = 2\phi^4$ exactly.
3. $p_1/p_0 = \phi^4/2$ exactly.
4. $3\sqrt{p_j} \in \{\sqrt{2}, \phi^2, 1/\phi^2\}$ exactly.
5. $74 \times \phi^8$ matches $m_\tau/m_e|_{\theta=2/9}$ to 0.030 %.
6. $30 \times \phi^4$ matches $m_\mu/m_e|_{\theta=2/9}$ to 0.56 %.
7. All three ratios carry only ϕ -powers (no π , no $\cos$).
8. $\varepsilon \approx 27\xi/12$ (0.49 % match).
9. $74 \times \phi^8 \times (1 + 27\xi/12)$ matches m_τ/m_e to 0.00015 %.

5 Galois Structure of ε : Observation and Limit

Numerical observation

The deviation ε from Section 3 lies close to a Galois-native ξ -expression:

$$\varepsilon \approx \frac{27\xi}{12} = \frac{|GF(27)| \cdot \xi}{n_{\phi,\mu} \cdot n_{\theta,e}} = \frac{3}{10000}, \quad \text{dev. 0.49 \%} \quad \mathbf{[K]} \quad (8)$$

All three factors – $27 = |GF(27)|$, $\xi = 4/30000$ and $12 = n_{\phi,\mu} \times n_{\theta,e} = 4 \times 3$ – are already known with status **[B]** from Doc. 338. Substituting:

$$\frac{m_\tau}{m_e} \approx 74 \times \phi^8 \times \left(1 + \frac{27\xi}{12}\right) \quad (0.00015 \%) \quad \mathbf{[K]} \quad (9)$$

Limit of the analytical proof path

A proof that $\varepsilon = 27\xi/12$ holds *exactly* cannot be established via the FFGFT formula. The test attempt:

From Doc. 352: $m_\tau/m_e(\text{bare}) = (25/12) \cdot \xi^{-5/6}$ (from winding numbers $r_\tau/r_e = 25/12$, $p_\tau - p_e = -5/6$), with fraktal correction coefficient 117.4ξ . A linearisation $(25/12) \cdot \xi^{-5/6} \approx 74 \times \phi^8 \times (1 + c_{\text{bare}} \cdot \xi)$ yields $c_{\text{bare}} \approx 119.1$.

The net coefficient is:

$$c_{\text{bare}} - 117.4 \approx 1.72 \neq \frac{27}{12} = 2.25 \quad (23 \% \text{ discrepancy})$$

The approach fails for two reasons:

1. $\xi^{-5/6}$ is not a polynomial in ξ : the expansion of $(25/12) \cdot \xi^{-5/6}$ around the fixed point $74 \times \phi^8$ yields a power series in $\xi^{1/6}$, not in ξ .
2. The coefficient 117.4 in Doc. 352 is itself numerical **[K]** (rounded value); both sides carry uncertainties that add up to $\sim 23 \%$.

Status. The identification $\varepsilon \approx 27\xi/12$ is a numerical observation **[K]** (0.49 % deviation). All factors are Galois-native from Doc. 338. An exact analytical derivation – showing why the net coefficient is precisely $27/12$ – requires a perturbation expansion in $\xi^{1/6}$ that goes beyond the current state of the corpus. This step remains **[S]**.

6 Summary: What Is Computable from the p -Values

Statement	Status	Source
$p_1/p_2 = \phi^8, p_0/p_2 = 2\phi^4$	[B]	A_5 geometry, Doc. 293
$3\sqrt{p_j} \in \{\sqrt{2}, \phi^2, 1/\phi^2\}$	[B]	A_5 geometry, Doc. 293
$m_\tau/m_e \approx 74 \times p_1/p_2$	[K]	Galois factor 74, Doc. 338
$m_\mu/m_e \approx 30 \times p_0/p_2$	[K]	Galois factor 15, Doc. 338
$\varepsilon \approx 27\xi/12$	[K]	Galois+ ξ , Doc. 338
Derivation of factors 74, 30 from p -values	[S]	open
Analytic proof $\varepsilon = 27\xi/12$ exactly	[S]	open

What follows directly from the p -values. The p -values p_0, p_1, p_2 from the A_5 geometry (Doc. 293) deliver exactly the ϕ -skeleton of the lepton masses: the ratios $p_1/p_2 = \phi^8$ and $p_0/p_2 = 2\phi^4$ are purely geometric identities with no free parameter.

What requires additional input. Converting to numerical mass ratios requires the Galois factors 74 and 30. These come from the $GF(3^n)$ structure of Doc. 338, not from the p -values themselves. The direct route

$$p_0, p_1, p_2 \xrightarrow{\text{exact}} \phi\text{-skeleton} \xrightarrow{\text{Galois, Doc. 338}} m_\tau/m_e, m_\mu/m_e$$

is fully described, but the second step is observation, not derivation. A fully direct route – in which the Galois factors themselves follow from the $A_5/\mathbb{Z}_3$ structure – is the remaining open edge.

What is genuinely new. Prior to this document, Doc. 293 had computed the p -values but had not identified their ratios as exact ϕ -powers. The identities $p_1/p_2 = \phi^8$ and $p_0/p_2 = 2\phi^4$ are new. The observation $m_\tau/m_e \approx 74 \times (p_1/p_2)$ connects, for the first time, the A_5 geometry of Doc. 293 to the Galois structure of Doc. 338 through a concrete mass ratio.